Active Learning for Sparse Neural Encoding in Macaque AIT Cortex: A Virtual Testbed, Novel Sampling Algorithms, and Reinforcement Learning Meta-Policy

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Abstract

Neurons in the macaque anterior inferior temporal (AIT) cortex exhibit extreme sparse encoding, responding selectively to only a handful of natural images among vast candidate pools. This poses a fundamental challenge for active sampling: how can we efficiently discover neuronal preferences when the vast majority of samples elicit near-zero responses? We address this challenge through three contributions. First, we construct a biologically plausible virtual testbed for sparse AIT neurons using the Maximum Entropy Principle (MEP), replicating real neural statistics without ad hoc micro-mechanism assumptions. Second, we propose two novel active sampling algorithms—Response-Stratified Fisher Sampling (RSFS) and Mean Collapse Escape Sampling (MCES)—that explicitly counter the "mean collapse" phenomenon where surrogate models collapse to near-zero predictions under sparse regimes. Third, we formulate active sampling as a Markov Decision Process and train a reinforcement learning (RL) meta-policy that adaptively switches among sampling strategies, establishing an empirical upper bound on linear decoding performance. Experiments on our virtual testbed demonstrate that RSFS and MCES significantly outperform random sampling (Wilcoxon $p < 0.001$), while the RL meta-policy dynamically orchestrates strategy transitions from error-sensitive exploration to geometric coverage and mean-collapse avoidance, closely tracking the oracle upper envelope across all training stages.

1 Introduction

The primate ventral visual pathway transforms pixel-level inputs into semantic object representations through hierarchical processing from V1 → V2 → V4 → IT [Felleman and Van Essen, 1991, Tanaka, 1996]. At the apex of this hierarchy, the anterior inferior temporal (AIT) cortex contains neurons with extremely sparse and selective tuning: each neuron responds vigorously to only a small fraction of natural images [Tsao et al., 2006, Freiwald et al., 2009]. This sparse encoding strategy is computationally efficient [Olshausen and Field, 1996], yet it creates a severe sampling dilemma for experimental neuroscience.

Recent large-scale electrophysiology using Neuropixels 2.0 probes has revealed that as neuronal sparsity increases, the predictive performance of deep neural network (DNN) models degrades significantly [Yamins et al., 2014, Zhuang et al., 2021]. A critical question remains unanswered: does this predictive failure stem from fundamental representational limitations of current models, or from insufficient coverage of the neuron's narrow preference manifold under passive random sampling? If the latter, then active sampling—algorithmically selecting the most informative stimuli—could address the entanglement between "model representational failure" and "data sampling insufficiency."

However, designing active sampling algorithms for AIT sparse encoding faces three unique challenges: **(C1)** *Mean collapse*: when most samples elicit zero response, surrogate models trained on randomly sampled data collapse to near-zero predictions, losing all sensitivity to rare high-response stimuli; **(C2)** *High-dimensional search*: AIT neurons possess near-global receptive fields, precluding spatial dimensionality reduction strategies effective in early visual areas [Op de Beeck and Vogels, 2000]; **(C3)** *Dynamic exploration–exploitation trade-off*: the optimal sampling strategy evolves as the model's knowledge of the feature space grows, requiring adaptive meta-level scheduling.

We address these challenges through three main contributions (Figure 1). First, we construct a **virtual sparse AIT neuron testbed** using the Maximum Entropy Principle (MEP) [Schneidman et al., 2006], constrained by sparsity distribution, effective dimension, and response decorrelation—matching real AIT statistics without introducing unverified micro-mechanisms. Second, we propose **RSFS** and **MCES**, two active sampling algorithms that counter mean collapse through response-space stratification and mean-escape penalties respectively. Third, we frame sampling as an MDP and train an **RL meta-policy** (DQN pre-training → PPO fine-tuning) that adaptively switches among strategies, establishing the empirical upper bound of linear decoding. Our code and data will be released upon acceptance.

2 Related Work

Computational proxies for visual cortex. Task-driven deep convolutional neural networks (DCNNs) trained on object classification spontaneously develop representations that linearly predict neural responses across the ventral stream [Yamins et al., 2014, Yamins

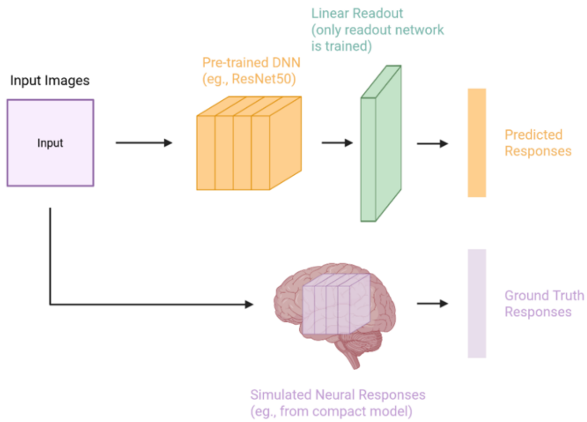


Figure 1: Dual-DNN virtual simulation framework. A pretrained "virtual brain" (ResNet-50) generates ground-truth responses; a readout network (ViT features + linear ridge regression) is trained via active sampling to predict these responses.

and DiCarlo, 2016]. Self-supervised models (SimCLR, MoCo) further improve natural scene predictions without human labels [Zhuang et al., 2021]. These models serve as "computational proxies" [Doerig et al., 2023] for probing high-level visual encoding.

Active sampling in neuroscience. Early work by Foldiak [Földiák, 2001] and Lewi et al. [Lewi et al., 2009] used closed-loop systems to identify preferred V1 stimuli. DiMattina and Zhang [DiMattina and Zhang, 2013] systematized adaptive stimulus optimization into three paradigms: optimal stimulus search, iso-response mapping, and system identification. Cowley et al. [Cowley et al., 2022] proposed population-level adaptive stimulus selection ("Adept") for V4, later extending to compact DNN models [Cowley et al., 2026]. High-contrast "gaudy" images were found to accelerate DNN training for visual cortex modeling [Cowley et al., 2020].

Active learning in machine learning. Bayesian experimental design [Chaloner and Verdinelli, 1995] provides principled criteria (A-, D-, E-optimality) for sample selection. For deep learning, uncertainty-based [Gal and Ghahramani, 2016] and diversity-based [Ash et al., 2020] methods are dominant. However, these methods assume relatively continuous target distributions, making them ill-suited for extreme sparsity where uncertainty estimates themselves become unreliable.

RL for adaptive experiment design. Reinforcement learning has been applied to automate experimental design in chemistry [Shiri et al., 2022] and neuroscience [Golub et al., 2018], but meta-level scheduling of multiple active sampling strategies for sparse neural encoding remains unexplored.

3 Methods

3.1 Virtual Sparse AIT Neuron Testbed

Feature extraction. We use pretrained ResNet-50 (layer3), VGG-19 (conv5_4), DenseNet-121 (denseblock3), ViT-Base (5th encoder), DINOv2-Base (8th block), and MAE (8th block) as feature encoders. Features are standardized, then projected to 1000 dimensions via PCA, forming the input space $\mathbf{X} \in \mathbb{R}^{1000}$.

Shifted ELU activation. To simulate biological baseline firing under extreme sparsity, we introduce a shifted exponential linear unit:

$$r = \alpha \cdot (\mathrm{ELU}(\mathbf{X}_{\mathrm{norm}}\mathbf{W} + \mathbf{b}) + 1) + \epsilon, \tag{1}$$

where $\mathrm{ELU}(x) = x$ if $x \geq 0$ else $\alpha_0(e^x - 1)$, $\alpha = 0.03$ controls baseline thickness, and ϵ is a small constant. This ensures smooth, non-zero tail decay mimicking spontaneous firing.

Maximum entropy generation. Rather than Gibbs sampling from a Boltzmann distribution (prohibitively slow in high dimensions), we directly minimize a composite energy function via gradient descent. The energy function enforces:

$$\mathcal{L}_{\mathrm{sparsity}} = \sum_i \omega_i (S_i - S_i^*)^2, \tag{2}$$

$$\mathcal{L}_{\mathrm{pop}} = (\bar{r} - \bar{r}^*)^2 + (\sigma_r - \sigma_r^*)^2, \tag{3}$$

$$\mathcal{L}_{\mathrm{ED}} = (\mathrm{ED} - \mathrm{ED}^*)^2, \tag{4}$$

$$\mathcal{L}_{\mathrm{VB}} = \frac{1}{N} \sum_i (v_i - \bar{v})^2, \tag{5}$$

$$\mathcal{L}_{\mathrm{curve}} = \sum_k \tau_k (r_{(k)} - r_{(k)}^*)^2, \tag{6}$$

$$\mathcal{L}_{\mathrm{DC}} = \mathrm{MSE}(\mathrm{CDF}(\rho_{ij}), \mathrm{CDF}(\rho_{ij}^*)), \tag{7}$$

where S_i is lifetime sparseness $S = \frac{1-(\sum r_i/n)^2/(\sum r_i^2/n)}{1-1/n}$; ED is effective dimension $\mathrm{ED} = (\sum \lambda_i)^2 / \sum \lambda_i^2$; ρ_{ij} are pairwise Pearson correlations; and * denotes target statistics from real AIT data. The total loss $\mathcal{L} = \sum_k \lambda_k \mathcal{L}_k$ is optimized with AdamW for ∼80k iterations.

3.2 Mathematical Framework

Problem formulation. Let $\mathcal{U}$ be the candidate pool of N natural images. At round t, we have labeled dataset $\mathcal{D}_{t-1}$, readout model M_{t-1}, and unlabeled pool $\mathcal{U}_t = \mathcal{U} \setminus \mathcal{D}_{t-1}$. We select a batch $\mathcal{B}_t \subset \mathcal{U}_t$ of size N_{sampling} to maximize the population mean decoding performance.

Acquisition function. A generic acquisition function $a(\mathbf{x}; M_{t-1}, \mathcal{D}_{t-1})$ maps each candidate $\mathbf{x} \in \mathcal{U}_t$ to a scalar score quantifying its information value. The active sampling objective is to choose the subset with the highest aggregate score:

$$\mathcal{B}_t^* = \arg \max_{\mathcal{B} \subset \mathcal{U}_t, |\mathcal{B}|=N_{\text{sampling}}} \sum_{\mathbf{x} \in \mathcal{B}} a(\mathbf{x}). \tag{8}$$

For a linear ridge regression readout with regularization λ, the posterior covariance admits a closed form. Define the information matrix after observing $\mathcal{D}_{t-1}$ as

$$\mathbf{S}_t = \lambda \mathbf{I} + \mathbf{X}_t^\top \mathbf{X}_t, \tag{9}$$

where $\mathbf{X}_t$ stacks the features of all selected samples. Under the D-optimal design criterion, the information gain for adding candidate $\mathbf{x}$ is

$$a_{\text{D-opt}}(\mathbf{x}) = \mathbf{x}^\top \mathbf{S}_t^{-1} \mathbf{x}, \tag{10}$$

which measures the reduction in parameter uncertainty. In practice, we maintain $\mathbf{S}_t^{-1}$ via the Sherman-Morrison formula, reducing updates from $\mathcal{O}(d^3)$ to $\mathcal{O}(d^2)$.

3.3 Active Sampling Algorithms

We implement four concrete acquisition strategies within the above framework. All experiments use mixed sampling: 50% of each batch is selected by the active algorithm and 50% is random, ensuring baseline exploration.

Global Greedy (baseline). This strategy greedily maximizes Eq. (10) at each step:

$$a_{\text{GG}}(\mathbf{x}) = \mathbf{x}^\top \mathbf{S}_t^{-1} \mathbf{x}, \tag{11}$$

which measures marginal information gain in feature space. We implement it with dynamic rolling subsets for tractability on the 100k candidate pool.

Response-Stratified Fisher Sampling (RSFS). RSFS addresses challenge C1 by enforcing response-space diversity. It operates in three stages: (1) *Response stratification*: spherical K-means clusters response vectors by direction; quantile binning partitions by magnitude, creating a grid of response buckets. (2) *Water-filling quota allocation*: buckets with fewer historical samples receive higher quotas, approximating entropy maximization. (3) *Feature-space scoring*: within the selected bucket b^*, the highest Fisher information sample is chosen:

$$a_{\text{RSFS}}(\mathbf{x} \mid b^*) = \mathbf{x}^\top \mathbf{S}_t^{-1} \mathbf{x}, \quad \mathbf{x} \in \mathcal{U}_{b^*}. \tag{12}$$

Mean Collapse Escape Sampling (MCES). MCES explicitly penalizes mean collapse via a dual-metric score. Let $\boldsymbol{\mu}_{\text{train}}$ and $\boldsymbol{\sigma}_{\text{train}}$ be the mean and std of predictions on the current training set. The adaptive sensitivity is $\alpha = 1.0/(\boldsymbol{\sigma}_{\text{train}} + \epsilon)$. The acquisition function is:

$$a_{\text{MCES}}(\mathbf{x}) = \underbrace{(\mathbf{x}^\top \mathbf{S}_t^{-1} \mathbf{x})}_{\text{geometric uncertainty}} \cdot \underbrace{\exp(-\alpha \|\mathbf{y}_{\text{pred}} - \boldsymbol{\mu}_{\text{train}}\|_1)}_{\text{mean collapse penalty}}. \tag{13}$$

This nonlinear fusion dynamically trades off feature-space orthogonality against response-space deviation from the collapsing mean.

3.4 RL Meta-Policy for Adaptive Sampling

We formulate active sampling as an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P})$.

State space. $\mathbf{s}_t \in \mathbb{R}^5$ comprises: (1) parameter uncertainty $\mathbf{S}_t = \lambda \mathbf{I} + \mathbf{X}_t^\top \mathbf{X}_t$ (information matrix trace), (2) prediction variance σ^2, (3) sampling progress ρ, (4) performance trend Δr, and (5) response distribution statistics (μ_r, σ_r).

Action space. $\mathcal{A} = \{\text{Random, Global Greedy, RSFS, MCES}\}$.

Reward. We use delta reward to mitigate credit assignment:

$$R_t = r_t - r_{t-1}, \tag{14}$$

where r_t is the population mean Pearson r after round t. The discount factor $\gamma = 0.6$ prioritizes short-term adaptation.

Two-stage training. *Stage 1 (DQN offline pre-training)*: We generate 100k transitions via uniform random action selection, then train DQN:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}}[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2]. \tag{15}$$

Stage 2 (PPO online fine-tuning): PPO directly optimizes $\pi(a|s)$ with clipping to avoid destructive updates:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \hat{A}_t, \text{clip}(\cdot, 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]. \tag{16}$$

4 Experiments and Results

4.1 Virtual Testbed Validation

We first validate that our MEP-based virtual testbed faithfully reproduces real AIT statistics across three levels (Figure 2):

Single-cell distributions. Ranked firing rate curves show that virtual neurons (all backbone networks) replicate the high selectivity and smooth long-tail decay of real AIT neurons, unlike standard ReLU which exhibits hard truncation artifacts.

Population topology. Effective dimension (ED) distributions from all models overlap with real AIT

($N = 40$ random subsamples; Mann-Whitney U, $p > 0.05$ for all). Neuron-neuron Pearson correlations cluster tightly around zero with Gaussian shape, confirming decorrelated population coding.

Sparsity alignment. Violin plots show median and distribution shape match real AIT (Mann-Whitney $p \in [0.141, 0.654]$). All 50-neuron populations pass non-parametric equivalence tests.

For downstream experiments, we adopt a **cross-architecture** protocol: ResNet-50 generates ground-truth responses (biological proxy), while ViT serves as the prediction readout. This deliberately introduces a representational gap, simulating real-world model-brain mismatch.

4.2 Active Sampling Performance

Decoding efficiency. Paired scatter plots (Figure 3) at sessions 2, 6, 10, 15 show all three active algorithms significantly outperform random sampling (Wilcoxon signed-rank, $p < 0.001$ at all stages). Global learning curves (Figure 4A) confirm this with non-overlapping 95% confidence intervals.

Mean collapse phenomenon. A striking metric inconsistency emerges: while active sampling achieves higher Pearson r, it shows *higher* RMSE and MAE than random sampling. This reveals mean collapse: random sampling's zero-response dominance causes the model to collapse to near-zero predictions, minimizing absolute error but destroying response-shape fidelity. Active sampling breaks this by injecting high-response "long-tail" samples, trading absolute precision for correlation-based shape recovery. The declining MAE/RMSE ratio (Figure 4D) confirms that active model error shifts from uniform background bias to rare-peak fitting fluctuations.

4.3 RL Meta-Policy Results

The RL meta-policy (purple, Figure 5) closely tracks the oracle upper envelope formed by the best fixed strategy at each session. This establishes that: **(i)** the current linear readout architecture has an empirical performance ceiling that active sampling can reach; **(ii)** no single fixed algorithm achieves this ceiling throughout; **(iii)** adaptive scheduling is essential.

Strategy evolution (Figure 6) reveals interpretable phase transitions:

- **Session 2 (cold start):** RSFS dominates (~66%). Error-sensitive exploration rapidly breaks underfitting.
- **Sessions 3–8 (coverage):** Global Greedy rises to ~48%. Geometric orthogonality expands the feature boundary.
- **Sessions 9–12 (trap escape):** MCES steadily increases. Mean-collapse avoidance becomes critical as the model enters the "zero-response plain."
- **Sessions 13–15 (refinement):** RSFS returns with random sampling (~20%). Marginal gains from active sampling diminish; random exploration prevents overfitting.

5 Discussion and Conclusion

We present a systematic framework for active learning of sparse neural encoding in the macaque AIT cortex. Our MEP-based virtual testbed provides a biologically grounded benchmark for evaluating sampling algorithms under extreme sparsity. The proposed RSFS and MCES algorithms address mean collapse through complementary mechanisms—response-space stratification and mean-escape penalties—both achieving significant gains over random sampling. Most importantly, the RL meta-policy demonstrates that no single strategy suffices; instead, dynamic scheduling across algorithmic phases is required to approach the empirical decoding upper bound.

Limitations. The sim-to-real gap involves unmodeled nonlinear dynamics, non-stationary biological noise, and potential distribution shift for RL transfer. Future work will deploy this system in *in vivo* closed-loop experiments using Neuropixels 2.0, with online RL fine-tuning. Additional directions include generative stimulus synthesis (GANs/diffusion models) to escape natural image library boundaries, and multi-unit activity (MUA)-guided pre-screening to reduce sorting latency.

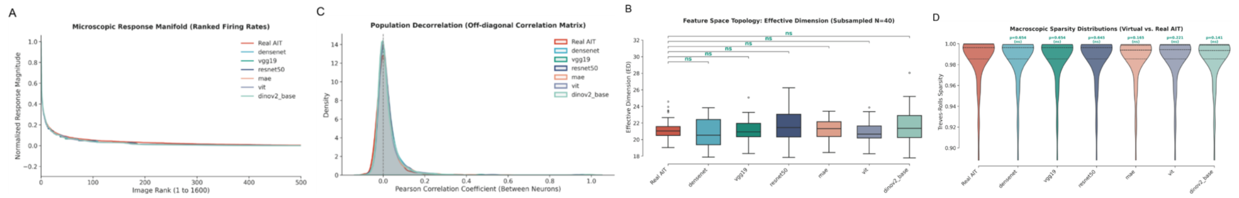
Broader impact. By establishing principled active sampling for sparse high-level visual encoding, this work provides both a theoretical framework and practical engineering pathway for accelerating neural characterization in the next generation of large-scale brain-computer interface experiments.

References

Jordan T Ash, Chicheng Zhang, Akshay Krishnamurthy, John Langford, and Alekh Agarwal. Deep batch active learning by diverse, uncertain gradient lower bounds. In *International Conference on Learning Representations*, 2020.

Kathryn Chaloner and Isabella Verdinelli. Bayesian experimental design: A review. *Statistical Science*, pages 273–304, 1995.

Benjamin R Cowley, Rachel C Williamson, Byron M Yu, Matthew A Smith, Adam Kohn, and Jonathan W Pillow. High-contrast gaudy images improve the training of deep neural network models of visual cortex. *Advances in Neural Information Processing Systems*, 33:1–12, 2020.



(a) Single-cell CDFs (b) Effective dimension (c) Decorrelation (d) Sparsity distribution

Figure 2: Virtual testbed validation. (A) Ranked firing rates show high selectivity + smooth tails. (B) ED matches real AIT (ns, Mann-Whitney). (C) Inter-neuron correlations peak at zero. (D) Sparsity distributions align ($p > 0.05$).

Benjamin R Cowley, Matthew A Smith, Adam Kohn, and Byron M Yu. Adaptive stimulus selection for optimizing neural population responses. *Advances in Neural Information Processing Systems*, 35:1–14, 2022.

Benjamin R Cowley, Petru Stan, Jonathan W Pillow, et al. Compact deep neural network models of the visual cortex. *Nature*, 652:947–954, 2026.

Christopher DiMattina and Kechen Zhang. Adaptive stimulus optimization for sensory systems neuroscience. *Frontiers in Neural Circuits*, 7:101, 2013.

Adrien Doerig, Ruben P Sommers, Katja Seeliger, Blake Richards, Jenann Ismael, Grace W Lindsay, Konrad P Kording, Talia Konkle, Marcel AJ van Gerven, Nikolaus Kriegeskorte, et al. The neuroconnectionist research programme. *Nature Reviews Neuroscience*, 24(7):431–450, 2023.

Daniel J Felleman and David C Van Essen. Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex*, 1(1):1–47, 1991.

Peter Földiák. Stimulus optimisation in primary visual cortex. *Neurocomputing*, 38:1217–1222, 2001.

Winrich A Freiwald, Doris Y Tsao, and Margaret S Livingstone. A face feature space in the macaque temporal lobe. *Nature Neuroscience*, 12(9):1187–1196, 2009.

Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, pages 1050–1059. PMLR, 2016.

Matthew D Golub, Patrick T Sadtler, Emily R Oby, Kieran M Quick, Stephen I Ryu, Elizabeth C Tyler-Kabara, Aaron P Batista, Steven M Chase, and Byron M Yu. Learning by neural reassociation. *Nature Neuroscience*, 21(4):607–616, 2018.

Jeremy Lewi, Robert Butera, and Liam Paninski. Sequential optimal design of neurophysiology experiments. *Neural Computation*, 21(3):619–687, 2009.

Bruno A Olshausen and David J Field. Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*, 381(6583): 607–609, 1996.

Hans P Op de Beeck and Rufin Vogels. Spatial sensitivity of macaque inferior temporal neurons. *Journal of Comparative Neurology*, 426(4):505–518, 2000.

Elad Schneidman, Michael J Berry, Ronen Segev, and William Bialek. Weak pairwise correlations imply strongly correlated network states in a neural population. *Nature*, 440(7087):1007–1012, 2006.

P Shiri et al. Accelerating materials discovery using machine learning, high-performance computing and robotics. *npj Computational Materials*, 8:248, 2022.

Keiji Tanaka. Inferotemporal cortex and object vision. *Annual Review of Neuroscience*, 19(1):109–139, 1996.

Doris Y Tsao, Winrich A Freiwald, Roger BH Tootell, and Margaret S Livingstone. A cortical region consisting entirely of face-selective cells. *Science*, 311 (5761):670–674, 2006.

Daniel LK Yamins and James J DiCarlo. Using goal-driven deep learning models to understand sensory cortex. *Nature Neuroscience*, 19(3):356–365, 2016.

Daniel LK Yamins, Ha Hong, Charles F Cadieu, Ethan A Solomon, Darren Seibert, and James J DiCarlo. Performance-optimized hierarchical models predict neural responses in higher visual cortex. In *Proceedings of the National Academy of Sciences*, volume 111, pages 8619–8624. National Acad Sciences, 2014.

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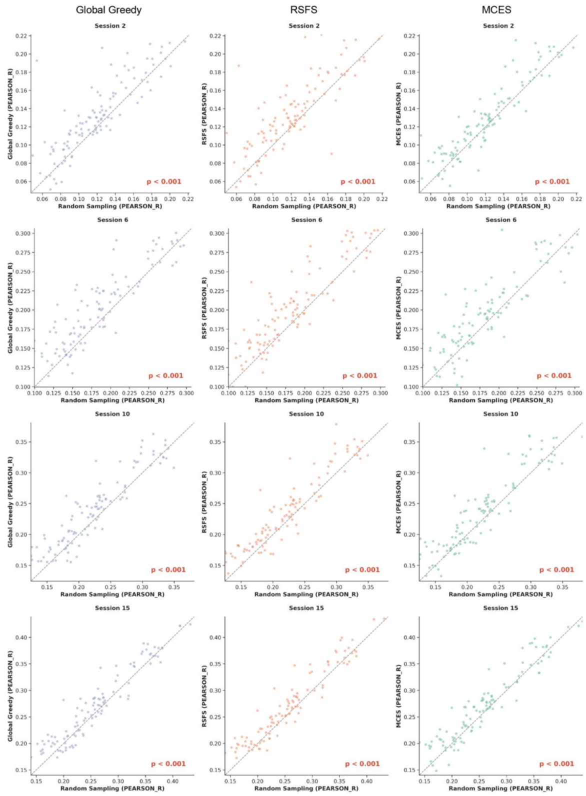

Figure 3: Paired scatter plots at sessions 2, 6, 10, 15. Each point is one virtual neuron. Points above the diagonal indicate active sampling outperforms random sampling ($p < 0.001$, Wilcoxon).

Chengxu Zhuang, Siming Yan, Aran Nayebi, Martin Schrimpf, Michael C Frank, James J DiCarlo, and Daniel LK Yamins. Unsupervised neural network models of the ventral visual stream. *Proceedings of the National Academy of Sciences*, 118(3): e2014196118, 2021.

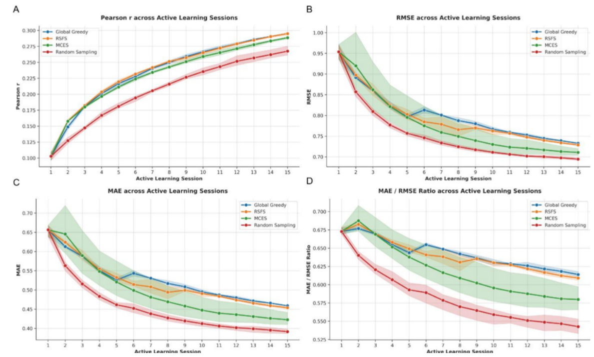

Figure 4: Global learning curves. (A) Pearson r: active sampling significantly exceeds random. (B-C) RMSE/MAE: active sampling higher due to mean-collapse escape. (D) MAE/RMSE ratio declines, showing error shifts to rare-peak fitting.

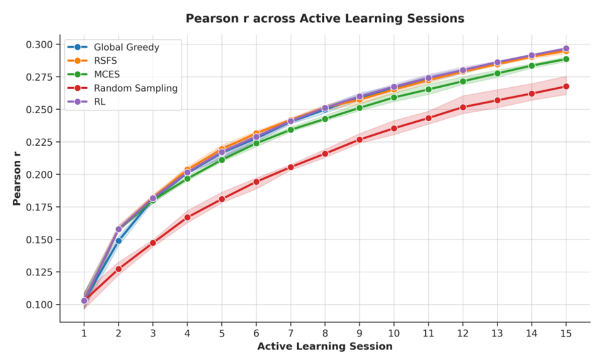

Figure 5: RL meta-policy vs. fixed strategies. The RL agent (purple) closely tracks the oracle upper envelope, demonstrating that adaptive scheduling is necessary to approach the empirical performance ceiling.

A Appendix

A.1 Lifetime Sparseness

Lifetime sparseness for a neuron with responses $\{r_1, \ldots, r_n\}$ to n stimuli:

$$S = \frac{1 - \frac{1}{n}\left(\sum_i r_i\right)^2 / \sum_i r_i^2}{1 - 1/n}. \tag{17}$$

A.2 Effective Dimension

For a population response covariance matrix with eigenvalues $\{\lambda_i\}$:

$$\mathrm{ED} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{18}$$

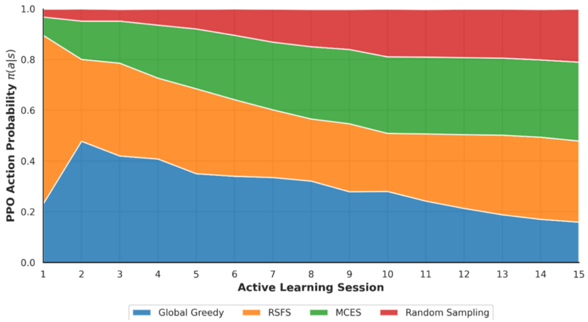

Figure 6: RL strategy evolution. Stacked area plot shows the probability $\pi(a|s)$ over 15 sessions. Clear phase transitions: RSFS (cold start) → Global Greedy (coverage) → MCES (trap escape) → mixed (refinement).

A.3 Sherman-Morrison Update

After selecting sample $\mathbf{x}$ with $\mathbf{S}_t^{-1}$ known, the update is:

$$\mathbf{S}_{t+1}^{-1} = \mathbf{S}_t^{-1} - \frac{\mathbf{S}_t^{-1}\mathbf{x}\mathbf{x}^\top\mathbf{S}_t^{-1}}{1 + \mathbf{x}^\top\mathbf{S}_t^{-1}\mathbf{x}}, \tag{19}$$

reducing complexity from $\mathcal{O}(d^3)$ to $\mathcal{O}(d^2)$.